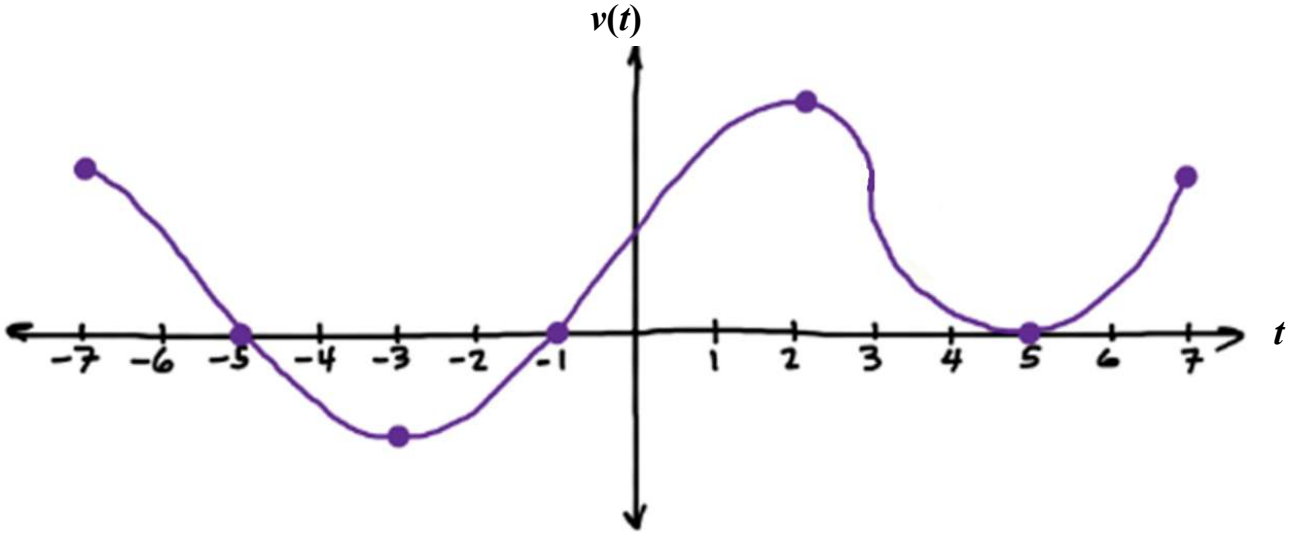
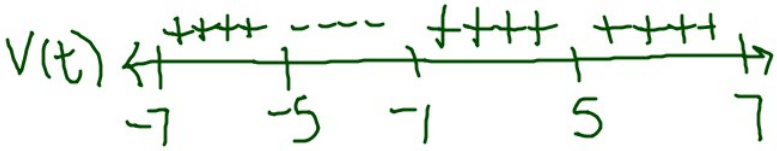
Warm-up Problem—Velocity
AP Calculus

Name SOLUTIONS

The graph of the velocity, v(t), of a particle on [–7, 7] is given below. The graph has horizontal tangent lines at x = –3, x = 2, and x = 5, and a vertical tangent line at x = 3.
(a) When is the particle moving to the right? to the left? JUSTIFY YOUR ANSWERS
(b) When is the acceleration positive? negative? JUSTIFY YOUR ANSWERS
(c) When is the particle speeding up? slowing down? JUSTIFY YOUR ANSWERS

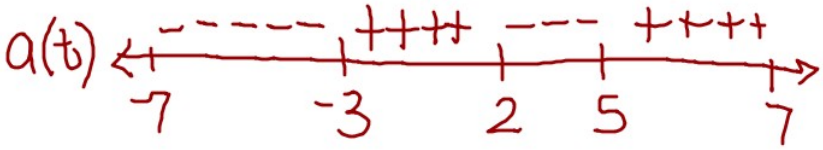


(a)



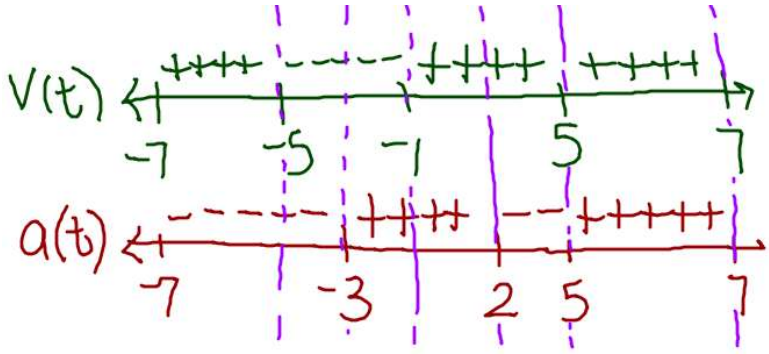
The particle is moving to the right on [–7, –5) ∪ (–1, 5) ∪ (5, 7] because $v(t) > 0$ on these intervals.
The particle is moving to the left on (–5, –1) because $v(t) < 0$ on this interval.

(b)



The acceleration is positive on (–3, 2) ∪ (5, 7) because $v(t)$ is increasing on these intervals.
The acceleration is negative on (–7, –3) ∪ (2, 5) because $v(t)$ is decreasing on these intervals.

(c)



The particle is speeding up on (–5, –3) because $v(t) < 0$ and v(t) is decreasing on this interval

The particle is speeding up on (–1, 2) ∪ (5, 7) because $v(t) > 0$ and v(t) is increasing on these intervals

The particle is slowing down on (–7, –5) ∪ (2, 5) because $v(t) > 0$ and $v(t)$ is decreasing on these intervals

The particle is slowing down on (–3, –1) because $v(t) < 0$ and $v(t)$ is increasing on this interval